

EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 03-301-1

Date: 14-5-2024

LOW VOLTAGE SINGLE CORE CABLES

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**SPECIFICATION FOR
SINGLE CORE SHEATHED
XLPE INSULATED CABLES 0.6/1(1.2) KV
ALUMINUM / COPPER, ARMoured AND
UNARMoured**

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1. SCOPE

This Specification covers the minimum technical requirement for design, engineering, manufacturing, construction, inspection, testing and supply of low voltage cables $U_0/U = 0.6/1$ kV, 50 Hz single core with stranded compact AL/CU conductor, extruded XLPE insulation armoured / Unarmoured and PVC over-sheath, the cables should be for (connecting between transformers and low voltage panels) underground direct buried with different cross-section areas suitable for direct burial, installation in ducts and for general purpose.

2. DEFINITIONS

a) Cross-linked polyethylene (XLPE):

XLPE is an acronym for cross linked polyethylene. XLPE is a thermosetting polymer meaning that the material (polyethylene (PE)) is cured under heat and in that process forms bonds in all directions forming a three dimensional matrix.

b) Polyvinyl chloride compound (PVC):

Combination of materials suitably selected, proportioned and treated, of which the characteristic constituent is the elastomer polyvinyl chloride or one of its copolymers. The same term also designates compounds containing both polyvinyl chloride and certain of its polymers

c) ATA

Aluminum Tape Armour (ATA) is used in single-core cables because it is non-magnetic. When an electric current passes through a cable it produces a magnetic field (the higher the voltage the bigger the field). The magnetic field will induce an electric current in steel armor (eddy currents), which can cause overheating in AC systems. The non-magnetic aluminum armor prevents this from happening.

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d) Rated voltage:

The rated voltage of a cable is the reference voltage for which the cable is designed and which serves to define the electrical tests

The rated voltage is expressed by the combination of two values U_0/U , expressed in volts:

U_0 being the r.m.s. value between any insulated conductor and "earth" (metal covering of the cable or the surrounding medium);

U being the r.m.s. value between any two-phase conductors of a multicore cable or of a system of single-core cables.

In an alternating current system, any rated voltage of a cable shall be at least equal to the nominal voltage of the system for which it is intended

This condition applies both to the value U_0 and the value U

Note: The operating voltage of a system may permanently exceed the nominal voltage of such a system by 10%. A cable can be used at a 10% higher operating voltage than its rated voltage if the latter is at least equal to the nominal voltage of the system

e) Test voltage:

Unless otherwise specified, the test voltages shall be a.c. 49 Hz to 61 Hz of approximately sine-wave form, the ratio peak value/r.m.s. value being equal to $\sqrt{2}$ with a tolerance of $\pm 7\%$. The values quoted are r.m.s. values.

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3. APPLICABLE CODES AND STANDARDS

Unless otherwise specified in this specification, the L.V cables should comply with the latest edition of IEC standard and ISO codes of standards as following:

Table (1)

IEC60502-1	Power cables with extruded insulation and their acc. For rated voltages from 1 KV ($U_m = 1.2$ KV, and 3 KV ($U_m = 3.6$ KV)
IEC60228	Conductors of insulated cables
IEC60811-100	Test methods for non-metallic materials / general
IEC60811-201	Test methods for non-metallic materials / general test
IEC60811-202	Test methods for non-metallic materials / general tests- measurements of thickness of non-metallic sheath
IEC60811-203	Test methods for non-metallic materials / general tests –measurements of overall dimensions
IEC60060	High-voltage test techniques
IEC60287	Calculation of the current rating

4. ENVIRONMENTAL CONDITIONS

The performance of the extruded XLPE insulated cables should be guaranteed for following environmental conditions, any differences in the guaranteed performance should be clearly set out in the offer.

Table (2)

Minimum ambient temperature	-5°C
Maximum ambient temperature	45°C (50 °C as option).
Maximum relative humidity	95%.
Maximum altitude	1000 m.
Maximum conductor temperature in normal use	90°C.

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5. SYSTEM DETAILS:

Frequency	50Hz
Number of phases	THEE PHASE
Nominal voltage	0.4 kV
Highest voltage	0.6 kV
Maximum S.C. temperature	250°C

6. DESIGN AND CONSTRUCTION REQUIREMENTS

6.1 General

6.1.1 Cables shall meet or exceed the requirements of this Specification in all respects. All cables shall be free of material and manufacturing defects, which would prevent it from meeting the requirements of this Specification.

6.2 Conductor

- The conductor shall consist of one of the following:
 - Copper or aluminum according to ...EDC requirements.
- The cables can be laid indoors, in air, in-ground, in ducts,
- The cables should class 2 plain AL or CU stranded wires. The conductors should be stranded compacted circular. The minimum number of wire strands in the conductor should be as following table.

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Table 3-Stranded conductor for single core.

Nominal C.S.A mm ²	Min NO of wires in the conductor		Max. DC resistance of conductor at 20°C Ω/km	
	Circular		CU. Conductor	AL conductor
Material	CU	AL		
50	19	19	0.387	0.641
70	19	19	0.268	0.443
95	19	19	0.193	0.320
120	37	37	0.153	0.253
150	37	37	0.124	0.206
185	37	37	0.0991	0.164
240	61	61	0.0754	0.125
300	61	61	0.0601	0.100
400	61	61	0.0470	0.0778
500	61	61	0.0366	0.0605
630	61	61	0.0283	0.0469
800	61	61	0.0221	0.0367

6.3 Insulation

Stranded cables insulation shall be extruded XLPE. Insulation shall comply with the appropriate requirements specified in IEC 60502-1.

The nominal minimum insulation thickness of stranded cables shall be according to relevant standard. The average insulation thickness shall not be less than the specified nominal value. Nominal Insulation thickness should be not less than the specified values given in the following table (4).

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Table 4-Stranded insulation for single core cable.

Nominal c.s.a mm ²	Thickness of insulation mm
50	1
70	1.1
95	1.1
120	1.2
150	1.4
185	1.6
240	1.7
300	1.8
400	2
500	2.2
630	2.4
800	2.6

The minimum thickness of the insulation at any point shall not fall below the nominal value by more than 0.1 mm + 10% of the specified nominal value

- Separation sheath (inner covering) in case of armour, the thickness of the separation sheath shall be as follow:

Cable C.S.A (mm ²)	Nominal (average) thickness of inner covering (mm)
1*50	1
1*70	1
1*95	1
1*120	1
1*150	1

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1*185	1
1*240	1
1*300	1
1*400	1.2
1*500	1.2
1*630	1.2
1*800	1.4

The measured smallest thickness of the PVC separation sheath, at any point, shall not fall by more than 0.2mm, below 80 % of the specified nominal value

Note: Inner covering thickness: mean value should be $\geq$ value corresponding to cable size in this table of above specifications, the lowest value of thickness at any point should not fall below table value by more than 20 % of this value + 0.2 mm

6.4 ARMOUR

- The armour should be applied on inner covering complying with of IEC60502-1.
- The armour should be ATA.
- The armour should be double tape armour of non-magnetic (aluminum).
- The thickness of tape should be derived from the following table:

Friction diam. Under the armour		Thickness of armour
Above	Up and including	
mm	mm	Mm
--	30	0.5
30	70	0.5
70	--	0.8

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- The tape armour shall be applied helically in two layers so that the outer tape is approximately central over the gap of the inner tape. The gap between adjacent turns of each tape shall not exceed 50 % of the width of the tape.

Cable C.S.A (mm ²)	Nominal thickness of armour (mm)
1*50	0.5
1*70	0.5
1*95	0.5
1*120	0.5
1*150	0.5
1*185	0.5
1*240	0.5
1*300	0.5
1*400	0.5
1*500	0.5
1*630	0.5
1*800	0.5

6.5 NON-METALLIC SHEATH

- The sheath shall be a PVC compound suitable for the cable operating temperature.
- The sheath should be applied directly over the armour.
- A suitable tape is permitted as a binder before application of an extruded over the sheath.
- The nominal thickness of the sheath should be not less than the values here in the following table (5)

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Table (5)

Cable C.S.A (mm ²)	Nominal (average) thickness of sheath (mm)	
	Copper / Aluminum	
	Armoured	Unarmoured
1*50	1.5	١,٤
1*70	1.6	1.5
1*95	1.7	١,٥
١*١٢٠	1.7	1.6
1*150	1.8	1.6
1*185	1.8	1.7
1*240	1.9	1.8
1*300	2	1.9
1*400	2.2	2
1*500	2.3	2.2
1*630	2.4	2.3
1*800	2.6	2.4

7 Cable lengths and drums

The cable length on each reel should be as follows:

Cable C.S.A (mm ²)	Cable length on reel (m)±5%	
	Copper	Aluminum
1*50	1000	1000
1*70	1000	1000
1*95	1000	1000
١*١٢٠	1000	1000
1*150	1000	1000
1*185	1000	1000
1*240	1000	1000

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1*300	1000	1000
1*400	500	1000
1*500	500	500
1*630	500	500
1*800	500	500

- All cable ends should be firmly secured to the reel and should be covered by heat-shrinkable end caps.
 - The cable reels should be covered with wooden slabs which tightened with steel tap of suitable thickness.
 - ...EDC has the right to change the cable length and consequently the reel diameter will be changed to max size 3 m.
 - The cables wooden drums shall be non-returnable.
 - The drum diameter shall not exceed 260cm width of cable drum including protruded fixing bolts shall not exceed 160cm.
- Cables end seals shall be installed in the factory to ensure that the cable remains watertight.
 - The drums shall be marked on both sides by cable type, size manufacturers / year of manufacture name ...EDC, length, Weight of cable, gross weight, and reel no.
 - On each reel the following data shall be printed on both sides:
 - * FOR --EDC ARAB REPUBLIC OF EGYPT
 - * LOW VOLTAGE CABLES
 - * MANUFACTURED BY:
 - * VOLTAGE CLASS: KV
 - * NUMBER OF CORES AND CROSS-SECTION mm2
 - * MATERIAL OF CONDUCTORS
 - * TYPE OF INSULATION:
 - * LENGTH OF CABLE: meters
 - * NET WEIGHT KGS
 - * GROSS WEIGHT: KGS
 - * REEL NO / YEAR OF MANUFACTURE

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8 Marking

The cable to which this specification relates shall be identified on the outer sheath with an identification of the manufacturer, the year of manufacture, the voltage 600/1000 v, and the ...EDC symbol.

Distance between marking should be not exceed one meter and each mark should serially indicate the remaining length of cable.

The cable to which this specification relates should be identified on the outer sheath with the following items (data):

- * Rated voltage
- * Size (number of cores*C.S. A of conductor mm²)
- * Material of conductor.
- * Insulation materials.
- * Manufacturer's name.
- * Year of manufacture.
- * Customer name (...EDC)
- * Cumulative cable length at every one meter.

9 TESTS

9.4 TEST CONDITIONS

- Voltage tests shall be made at ambient temperature $20 \pm 15^{\circ}\text{C}$ and other tests at ambient $20 \pm 15^{\circ}\text{C}$
- The frequency of alternating test voltage should be in the range 49–61 HZ

9.5 ROUTINE TESTS

1. ELECTRICAL RESISTANCE OF CONDUCTORS

- The measurement shall be made on all the conductors of each cable length.
- The complete cable length or sample shall be in the test room maintained at a reasonably constant temperature for at least 12 h before the test.

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- Alternatively, the resistance can be measured on a sample of conductor conditioned for at least 1h in a temperature-controlled oil bath.
- The measured value of resistance shall be corrected to a temperature of 20°C and 1 km length in accordance with the formula and factors in clause 5 of IEC 60228.
- The D.C. resistance of the conductor at 20°C shall not exceed the appropriate maximum value specified in IEC 60228.

2. H. VOLTAGE TEST

- The voltage test shall be made at ambient temperature, using either alternating voltage at power frequency or direct voltage.
- The test voltage shall be applied for 5 minutes between the conductor and the metallic screen.
- Test voltage:
 - The test voltage r.m.s shall be 3.5 kV.
 - When direct voltage is used, the applied voltage shall be 2.4 times the power frequency test voltage.
- Requirement: No breakdown of insulation shall occur.

9.6 TYPE TESTS

- Insulation resistance measurement at ambient temperature>
- Insulation resistance measurement at maximum conductor temperature in normal operation.
- Voltage test:

The core of the cable sample shall be immersed in water at ambient temperature for at least 1 hour before the test.

A power frequency voltage equal $4 U_0$ shall be gradually applied and maintained continuously for 4 hours between each conductor and the water.

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9.7 SPECIAL TESTS

i. CONDUCTOR EXAMINATION

- Compliance with the requirements of IEC 60228 shall be checked by inspection and by measurement when practicable.

II. MEASUREMENT OF THICKNESS OF INSULATION AND NON-METALLIC SHEATH

9.4.2.1 GENERAL

- Each cable length selected for the test shall be represented by a piece of cable taken from one end after having discarded, if necessary.
- If the average thickness measured or the lowest value measured fails to meet the requirements specified below two other pieces shall be checked.
- If both of the further pieces meet the specified requirements the cable is deemed to comply, but if either does not meet the requirements, the cable is deemed not to comply.

9.4.2.2 TEST METHOD

- The test method shall be in accordance with IEC 60811 as follows:

1. Measuring Equipment

A measuring microscope allowing a reading of 0.01 mm. instead of the microscope, a suitable measuring projector with a magnification power of at least 10 may be used. In the case of doubtful microscope shall be used.

- 2. Preparation of test pieces** any covering shall be removed from the insulation and the conductor shall be withdrawn, care being taken to avoid damage of insulation. Each test piece shall consist of a thin slice of insulation. The slice shall be cut with a suitable device (sharp knife or razor blade) along a plane perpendicular to the axis of the conductor. If a marking is stamped into the insulation or sheath thus giving rise to the local reduction of thickness the test piece shall be taken to include such marking.

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3.Measuring procedure

The test piece shall be placed under the measuring equipment with the plane of cut perpendicular to the optical axis.

- 1- When the inner profile of the test piece is a circle, six measurements shall be carried out radially equally spaced around the circumference.
- 2- When the insulation is taken from the conductor, six measurements are to be made radially in the positions where the insulation is thinnest i.e. between the ridges caused by strands.
- 3- If the inner substantially circular surface is not regular or smooth, the six measurements shall be made radially in the positions where the sheath is thinnest placing the cross-wire of the microscope.
- 4- The first measurement shall be made at the place where the insulation is thinnest.

9.4.2.3 REQUIREMENTS

A- INSULATION

For each piece of core the average of the measured values, rounded to 0.1 mm, shall not be less than the specified nominal thickness, and the smallest value shall not fall below the nominal value by more than $0.1 \text{ mm} + 10\%$ of the specified nominal value i.e. $t_m \geq t_n - (0.1 + 0.1t_n)$ in millimeters where:

t_m = minimum thickness

t_n = nominal thickness

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B- NON-METALLIC SHEATHS

The piece of sheath shall comply with the following:

- For a sheath applied on a smooth cylindrical surface of the insulation of a single core cable the average of the measured values, rounded to 0.1 mm shall not be less than the specified nominal thickness and the smallest value measured shall not fall below the specified nominal value by more than 0.2 mm + 20 % of the specified nominal value (i.e. $t_m \geq 0.8t_n - 0.2$).

IMPORTANT NOTE:

- The factory laboratory must be equipped with all modern instruments and equipments required to perform all routine and special tests of IEC 60502-1 and 60228 with the accuracies mentioned in them.
- Offers from tenderers having no such laboratories shall not be considered.
- The average thickness of insulation and non-metallic sheath shall not be less than the specified nominal value.
- Dimensions shall be read by digital and micrometric **instruments**.

10 Guarantee

- The supplier should guarantee the Cable against all defects arising out of faulty designed or workmanship or of defective materials for a period of 12 months from putting in service, or 18 months after delivery, whichever expires earlier.
- The good quality and manufacturing of the materials should be guaranteed.

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- Any parts which on account of poor quality of the materials, manufacturing defects or poor assembly, should be replaced or repaired in the shortest time possible and free of charge.

11 Drawing and Catalogues:

- All necessary drawings of the cable construction, Technical catalogues containing the technical data and reference standards applied to the cables and instructions for erection, operation, quality control procedure and maintenance should be submitted with the offer.
- Any tender not accompanied with clear and complete guarantee tables should be rejected.
- Type test report/certificate shall be submitted to EDC.

12 Guarantee Table

- The tenderer should fill in the attached guarantee tables.
- Any tender not accompanied with clear and complete guarantee tables should be rejected

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GURANTEE TABLE

A - TECHNICAL PARTICULARS:-	
Rated voltage	KV
Manufacturer's name	
Type of cable	
Standards specifications applied to the cable	
B –conductor data:-	
Number of cores/cable	
Nominal Cross-sectional area /core	mm ²
Shape of the conductor	
Number of strands per core/diameter of strand	
Cross-sectional area of each strand	mm ²
Guaranteed cross-sectional area / cond.	mm ²
Overall diameter of conductor	mm
Weight of conductor / km	kg
DC Resistance of conductor at 20°C / km	ohm
Temp. Coeff. of resistance	
Current carrying capacity	Amp
C- Insulation & Insulation strength:-	
Insulation class	KV
Material of insulation	
Thickness of insulation	mm
Min insulation resistance at 90 °C	MΩ.km
Max. allowable final temp. according to cable specification	°C
Test voltage between core and earth	
a) AC & Time	KV & min
b) DC & Time	KV & min

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E – Inner covering:-	
material of inner covering	
Thickness of inner covering	mm
D-armour	
No of layers	
Thickness of armour	mm
F-outer jacket	
Material of the outer jacket	
Thickness of outer jacket	mm
External diameter of cable	mm
Weight of cable/drum	kg
Length of cable/drum	m
Diameter of drum	m
Width of drum	m
Gross weight of charged drum	kg
Min. bending radius of cable	mm

I/we certify that the above-mentioned data are guaranteed for the offered cable.

Date:

Signature: